# GridFormLab Time-Step Engine: MATLAB Reference Record

A/B Measurement of the NE39 SG31 Reclose 160-s Scenario on the MATLAB Engine

Same-Tree Arm/B′-Pathology Design — the Measurement Record against which the C++ *flagship* deliverable will be verified

Historical reference anchor: 6708.725530 s (informational — not a divisor; see §2)

Report contract: *every number traceable to an artifact; anything unmeasured is printed as pending, never estimated*

In-house Power-Flow Project

14 October 2026

**Abstract**

This report is the **MATLAB reference record** for the GridFormLab time-step engine workstream on 2026-10-14: the validated measurement of the MATLAB engine's numerical behaviour and wall-time on the NE39 SG31 reclose 160-s scenario, which the forthcoming C++ flagship deliverable (full engine as a GUI application, installer-distributed) must match bit-exactly and beat on speed. It records the same-tree measurement design (Arm A options off, re-measured today; the historical 6708.725530 s run serves as an informational anchor only), the completed Arm A artifact, the Arm B′ opt-on pathology resolved by clean-machine per-phase evidence, the measured future optimization targets (Lane E zero-dwell trap; refinement zone), and the provenance of every number. A companion report will compare the C++ engine against this record — parity in the D5 frame plus the measured speedup — as the flagship's acceptance evidence. *Verified so far: `25/25` golden-vector checks on the C++ core, `35/35` cumulative checks with the layout converter, `62/62` on the first power-flow batch, `148/0` then `170/0` Python test suites; short-window Lane results (residual reuse* **2.2**× *at* tol = 0*, parallel FD groups* **2.53**×**/3.33**×*, MEX parity `4/4`); the Arm A same-tree measurement (1972.52 s, SUCCESS at the historical reclose instant); and the Arm B′ pathology resolved by clean-machine evidence.* All results are quoted with their measurement conditions and with the artifact that carries the number, in the provenance table at the end. In line with the acceptance rulings (D1–D5), every still-pending quantity appears as a clearly marked placeholder for data, never as a number, and speedup categories are never summed.

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# 1 Scope, sources, and the evidence contract

**What this report covers.** Two related workstreams share one acceptance question — *is the engine faster without changing the answers?*

1. **The port**: a standalone C++ core (`src/core/`, `src/pf/` in the port repository `D:\textbackslash 4-year-project\textbackslash GridFormLab-port`) replacing MATLAB-internal numerics behind a header-defined C API, plus a Python package (`python/gridform/`) that owns the canonical data exchange and the acceptance comparator.
2. **MATLAB-side optimization lanes** that stay inside the MATLAB tree: Lane A (residual value threading in the composite Newton step), Lane E (parallel forward-difference column groups via `parfor`, and a MEX+OpenMP FD kernel, not yet integrated into the production path).

**All speedups sit behind default-off switches.** Every optimization delivered by this workstream — residual value threading, parallel FD groups, the MEX kernel — ships with its option defaulting to `off`/`false`: production behaviour of the MATLAB tree is byte-identical until each switch is explicitly enabled, each switch carries a legacy fail-closed fallback, and none of the verified results below changes what production computes today.

**Measurement design: same-tree A/B; the historical number as anchor.** Full-window 160-s speed is measured as **Arm A (options off) vs Arm B (options on)** on one tree, one environment, one session — not by division against the historical number. The historical baseline reference (`docs/project/baseline\_ne39\_reclose\_160s.md`, verified 2026-10-13 by a three-way cross-check of `V\_bus`, `delta\_sg`, and `df`: NE39 SG31 reclose, 160-s horizon, wall clock 6708.725530 s = 1.8635 h, `production\_certified=false`) is an **informational anchor**: it was measured on a different tree and settings, and no speedup anywhere in this report is computed by dividing into it. The A/B 160-s measurement itself is pending (§7).

**Evidence contract (why this report looks the way it does).** A previous report in this project was written against a trajectory that had not finished running, with no placeholders, and the missing numbers were invented by omission. This report is built on the opposite contract:

- every number is copied from a disk-verifiable artifact or project document, and the provenance table (Section 10) maps each number to its artifact;
- every measured speedup states its *measurement window and conditions* in the same sentence as the number;
- anything not yet measured appears only through the marked box or the red pending measurement tag — a blank field is *not* filled with a plausible value;
- where an earlier note circulating in the team carries a number this report cannot verify (one case found, Section 5), the discrepancy is printed instead of silently choosing a side.

**Classification.** The baseline run keeps its recorded classification `PROJECT\_DERIVED\_TRUE\_TRIP\_RECL` with `production\_certified=0` and `hardware\_certified=0`; the port acceptance thresholds of Section 4 are the project's own decision record, not an external standard. Nothing in this report is a production or hardware certificate.

# 2 Baseline reference: what every arm is measured against

## 2.1 Run identification and options

The baseline artifact is `output/diagnostics/ne39\_reclose\_applied\_probe\_20261011\_090300\_` (documented branch state pinned at commit `f79bead`; the branch label itself has since been moved, so recovery is by commit only — a caveat recorded in the baseline document). Options are read from `request.mat`, which is authoritative over scripts on disk: `dt`=0.001, `dt\_min`=6.103515625e-07 (= 0.0025/4096), `dt\_max`=0.025, `dt\_max\_armed`=0.01, stepper `adaptive` with strict LTE, `reject\_limit`=16, `ne39\_trial\_max\_steps`=32000, `budget.max\_full\_evaluations`=4, `t\_end`=160, `post\_reclose\_mode\_reselection`=false, `rannacher\_n`=0. The run was executed on an Intel i5-14400F (10 cores/16 threads), 31.8 GB RAM, MATLAB R2026b (26.2.0.3386108).

## 2.2 Event chronology and headline counters

Table 1: Baseline event chronology (from the verified baseline document; "sample" is the index on the accepted 18,852-sample grid).

| Event | $t$ (s) | Sample idx | Note |
|---|---|---|---|
| sg_trip (SG31) | 20.000000 | 1605 | applied; nine GFM committed |
| fault_on (bus 16, $Z_f = j0.1$) | 60.000000 | 5400 | applied |
| fault_clear | 60.150000 | 5877 | applied |
| sg_on request | 100.000000 | 9576 | applied |
| **SG31 reclose** | **119.521920776** | 13493→13494 | `SUCCESS`, actual_close=true |
| horizon end | 160.000000 | 18852 | `NUMERICAL\_HORIZON\_COMPLETE` |

Table 2: Baseline counters (the historical anchor's reference column — an informational record from a different tree and settings; the A/B comparison of §7 measures its own pairs). Rejected steps are computed as attempts minus accepted, public accepted path only — the `rejects=` column of `progress.log` is a per-step, reset-on-accept counter and is never used.

| Counter | Baseline value |
|---|---|
| wall clock `elapsed\_s` | 6708.725530 |
| `ts\_accepted\_steps` | 9423 |
| `ts\_step\_attempts` | 9473 |
| cumulative rejected (attempts−accepted, public path) | 50 |
| `ts\_newton\_iterations` (total) | 64052 |
| min/mean/max | 3 / 6.80 / 73 |
| `event\_transactions` | 5 |
| `residual\_evaluations\_avoided` | (lane-A counter; not present in baseline) |
| FD group count / parallel mode / workers | n/a (serial path) |
| accepted samples (audit) | 18852 |
| energy error (`pu\,s`) | $8.98 \times 10^{-11}$ |
| right-limit KCL norm (pu) | $4.775 \times 10^{-10}$ |
| reclose guards $d\theta$ / $dV$ / $df$ | $0.4627^\circ$ / $1.33 \times 10^{-15}$ / $3.30 \times 10^{-6}$ |

The baseline document also records honest display notes carried into this report unchanged: strict_voltage_status=FAIL_ON_OBSERVED_SAMPLES (64 flagged samples), strict_nonvoltag and a private transition trial that reports STUDY_TRIAL_COMPLETE with refinement_samples=4292 and refinement_error=$5.52 \times 10^{-6}$ — the trial artifact carries *no* rejection counter, so no rejection statement of any kind is made about the private trial.

# 3 Port architecture

## 3.1 Components

| Layer | Location | Contents |
|---|---|---|
| C++ core | GridFormLab-port/src/core/ | gridform_fd_core.\{h,cpp\} (forward-difference kernel, mxArray shim removed), gridform_linalg.\{h,cpp\} (dense LU / solve / rcond), gridform_layout.\{h,cpp\} (CompositeLayout: 1-based active_indices in, 0-based CSR out), tests with golden vectors; built as a single gridform_core.dll |
| C++ PF batch | src/pf/ | gridform_pf_model.h, gridform_pf_prepare_case.\{h,cpp\} (bus 8/10/12-col and line 4/5/6/7-col parsers, validation throws as GFL_ERR_INPUT), gridform_pf_results.\{h,cpp\} (36-field, alias-based), pf helper tests |
| Python package | python/gridform/ | canonical/ (hexfloat.py bit-hex, reader.py %.17g CSV), parity/ (comparator.py, tier2.py — D5 thresholds transcribed verbatim with a THRESHOLD_PROVENANCE pointer at decisions_20261014.md), pfsolver/ (capability registry, routing, grouping, interfaces), tools/runner.py, tools/parity_check.py |
| Golden vectors | src/core/tests/golden/ | 14 binary dumps from the real MEX kernel on a synthetic 3-bus/2-SG case (fd, meta, residual, r0, f0, u, x0, z + solve A,b,meta,x + rcond); produced by scripts/build_core_golden.m reading the experiment-tree MEX binary as reference |

### 3.2 Data flow

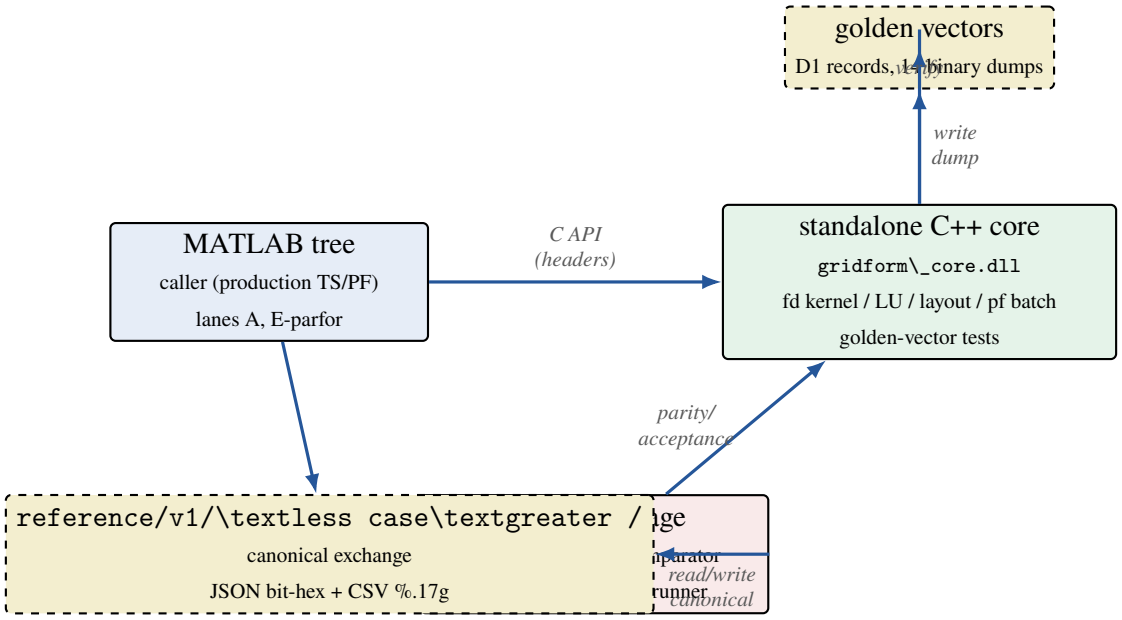



Figure 1: Port architecture. The MATLAB tree keeps production behaviour (optimization options default off); the standalone C++ core is verified against golden vectors dumped from the real MEX kernel; the Python package owns the D1 canonical exchange and the Tier-1/Tier-2 acceptance comparator and is the single place acceptance is decided. Dashed yellow nodes are data records, all of which carry per-record provenance.

### 3.3 Acceptance tooling behaviour (fail-closed)

The comparator `tools/parity\_check.py` implements the acceptance gate with three modes (`tier2`, `bit-exact`, `report-only`); a DECISION-field mismatch exits `1` with `DECISION\_MISMATCH\_NEEDS\_USER\_SIGNOFF` and cannot be washed away by a divergence-ledger id; non-finite data, shape or time-grid mismatch, and unmapped columns fail closed; every report carries per-input provenance. Peaks/settling comparisons are declared but deliberately unmapped until the canonical exporter schema exists — the engineer refused to guess units, and that refusal is recorded as correct behaviour, not an incomplete feature.

## 4 Parity methodology: the D1–D5 decision record

The acceptance framework is the project decision record (`GridFormLab-port/docs/decisions\_20261014`); its rulings are authoritative and are reproduced in abbreviated form here.

### 4.1 Canonical exchange (D1) — and its byte-reproducibility

Open records are exchanged as JSON with IEEE-754 bit-hex (`num2hex`) integers and CSV at %.17g precision; `.mat` is removed from the exchange. Every record must name its provenance (source file, field, commit, dirty state). This is what makes bit-level parity claims checkable by a third party.

The exchange format has one property the parity chain cannot work without: it is **byte-reproducible**. Verified by a two-run comparison on 2026-10-14 (port-scout selftest at 07:15

and 07:18, two full independent writes of the output directory): the same input produces byte-identical outputs,

| File | SHA-256 (identical across both runs) |
|---|---|
| `case.json` | `9614d639214c35f6aab22ff30ac43da14ed1ac53a7055eb1c9ce43c76f35a12f` |
| `events\_reference.json` | `b5e1ec1f6ac89e71e551dd0330ed3b4c3b055e101cd1e4a02c614ef41bc3cbe2` |
| `traj\_reference.csv` | `4af1628cb60988f1aea8ab2010252da56a16a5aeda3ee806c0831c9f09f90eb7` |

— no nondeterminism from field ordering or timestamps. Identical inputs yield identical bytes on any machine and in any round, which is exactly what cross-machine, cross-round parity comparison requires. Full hashes are reproduced with their provenance in §10.

## 4.2 Acceptance tiers (D5)

Table 3: D5 acceptance criteria. Tier 1 applies to pure optimization and refactoring merged back into the MATLAB tree; Tier 2 to the new engine, the port, and intended bug fixes. Thresholds are provisional under delegated authority and move only with evidence — never silently.

| Tier | Applies to | Criterion |
|---|---|---|
| 1 | optimization only (Lanes A/B/E, refactor) | **bit-identical, tol = 0, mandatory** for everything merged back to the MATLAB tree |
| 2 | new engine / port / intended fixes | DECISION fields (event order, actual trip/reclose times, `reclose\_status`, `failure\_id`, mode selection) **exact always**; continuous arrays on a common grid: $\lvert\Delta V\rvert \le 1 \times 10^{-4}$ pu, $\lvert\Delta f\rvert \le 1 \times 10^{-4}$ Hz, $\lvert\Delta\theta\rvert \le 0.01°$, $\lvert\Delta P\rvert/\lvert\Delta Q\rvert \le 1 \times 10^{-3}$ pu per device, peaks/settling $\le 1\%$; a verified bug-fix divergence goes into the divergence ledger and is not a failure, but any divergence touching a DECISION field needs user sign-off; over-threshold without explanation = fail, no self-service relaxation |
| speed target | this phase | $\ge 2\times$ on NE39 160 s (from the same record); a *target*, not a result — measured status is Section 7 |

## 4.3 Additional rulings D6–D9 (same record, later addendum)

- **D6 — `load\_model` canonical enum**: `cz\_p\_cz\_q` (alias `cz`) / `cc\_p\_cz\_q` / `cc\_p\_cc\_q`; any other value is a loud fail-closed error on every path. The new engine implements real semantics for every value; the MATLAB tree fixes only the lying default at `classical\_dae.m:244` (a declared zero-behaviour-change Tier-1 fix). All 18 audited runs that carry `request.mat` used `cz\_p\_cz\_q`, so no historical result diverges from this issue.
- **D7 — device_type collision (B3)**: the user's binding intent is "the reclose must succeed"; the behaviour that produces `reclose\_status=SUCCESS` at 119.521920776 s against the baseline may not be perturbed. Recorded as a known issue (`device\_type` shared between the 2-state and 7-state SG devices); the port replicates it as-is and any refactor touching it must first prove the reclose outcome identical (Tier 1).

- **D8 — `plot\_results` defaults**: the new engine defaults to `false` in every solver; the original MATLAB tree is not touched (display-only).
- **D9 — replicate, do not improve (B18/B20)**: the PF initial-state non-flat `angle\_spec\_deg` guess and the three distinct line-search budgets ($2^{-19}$ / $2^{-16}$ / lease 60) are parity-critical and are replicated exactly before any unification may even be proposed.

## 4.4 DECISION fields

The protocol fixes the fields that must match exactly in every arm: `converged`, `device\_modes\_history`, `actual\_reclose\_time`, `reclose\_status`, `actual\_mode\_reselection\_time`, `reselection\_status`, `requested\_sg\_on\_time`, `handback\_status`, `handback\_start\_time`, `handback\_duration\_s`, `handback\_complete\_time`, `failure\_id`, `committed\_config\_fingerpr` `pre\_event\_input\_fingerprint`, `selector\_table\_fingerprint`, `sg\_indices`. Counters are reported, never gated.

## 4.5 Standing prohibitions

Carried from the benchmark protocol and merge rules, unchanged: no residual/Jacobian reuse across different time, input, or limiter context; no change to the `jacobian\_evaluations` counter convention; every numerical path keeps a legacy fail-closed fallback; timing from cached replay is never counted as solver speedup; *do not* claim speedup equal to the core count — the MATLAB-side residual is still interpreted, so the ceiling argument of that kind is invalid by construction.

# 5 Verified results: port phases

Table 4: Port-phase verification status (all counts are passing checks in the respective suites; each row names its evidence — full pointers in Section 10).

| Phase | Scope | Checks | Evidence |
|---|---|---|---|
| P-C0 | Python harness | **148/0** | pytest run by the outgoing coordinator at harness close, exit 0 |
| P-C0Δ | parity-check suite added | **170/0** | `170 = 148 + 22` new `tests/test\_parity\_check.py`, verified by `python-dev` from the last pre-HOLD run; scope note: the Python suite is structural/unit coverage of the canonical format, registry, and comparator wiring — the numerical parity gate itself runs in MATLAB and C++ — so the count must not be presented as full numerical parity |
| P-C1 | C++ fd core, golden vectors | **25/25** | standalone runner `test\_golden\_vectors.cpp`: residual self-consistency, FD vs analytic ground truth, serial vs OpenMP bit-identity, linear algebra, layout converter — `CHECK` macros with no tolerance; reference build flags `-O2 -ffp-contract=off -fno-fast-math` locked by the shared `Makefile` parity contract |
| — | CompositeLayout converter | **35/35** | cumulative checks including the converter cross-check (fail-loud on indexing-base mismatch) |
| P-PF0 | PF batch 1 | **62/62** | core 35 + pf 27: helpers expression-verbatim (including the B18 non-flat initial guess), 36-field results, delta_idx fail-loud gate at `pf\_prepare\_case` |
| D1–D5 | decision record | folded | all rulings transcribed into the architecture doc and the comparator constants with provenance comments |
| Ledger | #D4-1 | **APPROVED** | intentional divergence recorded, text frozen |

**Number discrepancy reported, then resolved with evidence.** The original assignment brief for this report carried the figure "54 python tests". The engineer who owns the Python suite resolved this with static disk evidence: "54" is a stale count from *before* the test suites were ported into `GridFormLab-port/python` and predates the last audited states; the evidence-backed series is **148/0** at harness close (`pytest` exit 0 run by the outgoing coordinator) and **170/0** after the parity suite landed (148 + 22 in `tests/test\_parity\_check.py`), verified by `python-dev` on 2026-10-14 before the benchmark HOLD, with re-verification after the benchmark window assigned to `port-lead2`. A `.pytest\_cache/lastfailed` entry naming a single failed runner test was proven stale as well: the node id it records no longer exists in the code (the test was renamed when the runner behaviour changed), so the cache is a leftover of an intermediate run, not a standing failure. This report prints 148 and 170 with provenance; per the owner's correction, the 170 pytest *items* comprise 156 `def test\_` functions expanded by parametrization — the two counts are different units and are labelled as such wherever they appear.

**Not yet available at press time.** The MATLAB-side canonical exporter (the producer half of the D1 exchange) has no owner assigned yet, which stalls the end-to-end parity chain; PF golden vectors are deliberately on hold until the benchmark broadcast frees MATLAB and D1

provenance can be recorded. The `gfc\_*` ABI (P-C2) and its Python binding (P-C3) are queued behind PF batch close.

# 6 Verified results: MATLAB-side optimization lanes

Table 5: Measured optimization-lane results. Conditions are part of the result: all speedups are short-window or micro-benchmark measurements on an idle machine, and *none* of them is a NE39 160-s number (Section 7 is the only source for that). Speedup categories A (redundant work), E (parallelism), and G (early-phase cost) are never summed. Lane G's row is the measured pre-optimization target, not a speedup (§6.1).

| Lane / category | Result | Conditions |
|---|---|---|
| G — early-phase cost | target: **trip transaction (≥17 min, measured)** | |

All three new options ship switched off by design (merge rule: "defaults off, production behaviour unchanged"), each with a legacy fail-closed fallback, and all three respect the prohibition on cross-context residual reuse (the reuse wrapper is keyed on ($z$, closure token) with frozen and live limiter states kept apart).

## 6.1 Lane G — early-phase optimization: the measured target

Before any optimization of the early phase was designed, the workstream measured where the time actually goes — twice. The first measurement attributed the cost to the early window's per-step work; the second (profile round 2, killed at its gate) relocated the dominant block. Both are recorded here, because the second supersedes the interpretation — not the numbers:

Table 6: Early-window cost measurement (source: run `...054831\_227 raw.mat`; smoke schedule trip at $t$=2 s; all rates are measured values from the stored step record, except the per-iteration figure, which is derived by the annotated formula). **Both ratios are window-level attribution** — they include the one-shot trip transaction (≥17 min) inside the early window; the per-step interpretation is superseded (see Measurement 2), and the internal split is pending the transaction-timer profile.

| Window | $t \in (0.001, 2]$ s | $t \in (2, 25]$ s |
|---|---|---|
| accepted steps | 82 | 1,752 |
| Newton iterations (mean) | 3.0 | 6.75 |
| accepted step size $\Delta t$ | $2.5 \times 10^{-2}$ s (at `dt\_max`) | $2.5 \times 10^{-2}$ s |
| wall time per step | ~**32 s** (≈44.3 min total) | ~**0.51 s** (≈15 min total) |
| per-step cost ratio | **62**× *measured* — window-level attribution | |
| per-Newton-iteration ratio | ~140× *derived*: $62 \times (6.75/3) = 139.5$ — includes the one-shot transaction | |

**Status of these figures (attribution superseded, numbers kept):** the 62× per-step and ~140× per-iteration ratios are genuine measurements of the two windows — they stand as *window-level attribution*. They were initially read as the early window's residual/FD cost per step; profile round 2 relocated the dominant block (below), so these numbers are no longer claimed as a per-residual-call cost or as the Lane G target magnitude — only as what the windows cost, which remains true.

### 6.1.1 Measurement 2: the trip transaction finding

Profile round 2 (`profile\_lane\_g1\_20261014\_074637`, killed at its 20-minute gate, `KILLED.txt` in-folder) found the time's true address: the wall-clock block is not spread across early *steps* — it concentrates in a single **trip/commitment transaction** (selector authentication, transaction-scoped equilibrium work, right-limit Newton solve):

- the pre-trip steps are fast and normal: the first progress lines ($t = 0.001 \rightarrow 0.081$, then $t = 0.1000$ with $\Delta t = 1.9 \times 10^{-2}$ s) advance immediately for every simulated 0.01 s;
- at $t = 0.1000$ the run enters the trip transaction and produces *no further progress* for ≥ 17 minutes until the kill gate (08:04:41);
- the "slow early phase" of measurement 1 is therefore window-level bookkeeping: on the real 160-s schedule the same transaction sits at $t = 20$ s, where its wall cost surfaces in exactly the same way.

**Lane G target (revised — target pending transaction-split).** The dominant optimization target is the **trip/commitment transaction**. A combinatorial GFM-search budget may contribute inside its selector-authentication path — the split is not yet measured, because no transaction-internal profile percentages exist yet, so the Lane G target stands provisionally as "the trip transaction, split pending" and no per-component target is set. The per-step `residual`/FD path remains a legitimate Lane G sub-target, but it should *not* be expected to remove the 44-minute block: the block lives in the transaction. The transaction-timer profile (approved — timer-callback dump every 30 s, so a gate-kill still yields data) is the next instrument; until its numbers arrive this report states the target without a split:

| trip-transaction internal split: pending measurement — transaction-timer profile pending (item (d)); no percentage exists yet and none is estimated |
|---|

**What this closes — by counters.** Unchanged by the relocation: the early window is *not* a step-control pathology and *not* a rejected-step loop — `rejection\_history=` 0, `subdivision\_depth=` 0, and both selector/SSSSA trial counters = 0 across the completed run, and the step size sits at `dt\_max` throughout (fewer, bigger steps). The cost is real numerical work; the new finding locates it inside the transaction.

**What this report does *not* claim.** The iso arm measurement is run-to-run inside one smoke horizon; the *historical* 2026-10-11 baseline cannot be decomposed into early-phase vs remainder from its artifact (no per-line timestamps in `progress.log`), so this report never states how much slower the old tree was than the 11-October number — only that the early-heavy behaviour is the scenario's own, replicated without WIP by the iso arm. The final Lane G speedup claim waits for its A/B pair, and the transaction's internal split waits for the transaction-timer profile (item (d)).

**Ledger of the road here (three run episodes).**

1. `...053203\_494`: harness smoke with an over-long horizon ($t_{\text{end}} = 21 <$ `fault\_clear=` 60.15) — the engine's event-validation contract refused it fast, cleanly, and by design (`stability:ibr\_event\_schedule:badOrdering`); *not* a defect, an instrument working.

2. `...053450\_820`: killed at ~13 min before the `progress\_every` semantics were fully internalized; the forensic flat-line record (799 s, single progress line) led directly to the run-id-guard and inference rules this report now follows.

3. `...054831\_227`: completed (3,589.996 s for a 25- s smoke horizon) — the run that produced the window measurement above and the A/B noise verdict (0.8% against the iso pair, §7).

**Profile-round ledger.**

- profile round 2 — `profile\_lane\_g1\_20261014\_074637`, killed at the 20-minute gate (`KILLED.txt`): the kill that found the target. Artifacts: `request.mat`, `progress.log` (5 lines, last at $t = 0.1000$), `KILLED.txt`.

**Disowned values, family register.** Every `speedup\_vs\_baseline=...` line printed by a smoke-horizon summary belongs to a family of meaningless ratios and is disowned as a class: `10679.3764x` (run 1: crashed-derived), `1.8687x` (run 3: 25-s horizon vs 160-s baseline), and `3.42x` (the same mismatch in an early draft). The earlier `250--500$\times $` per-residual ratio is withdrawn separately: it was an inference from window data that the trip-transaction finding superseded, removed from this report entirely by the coordinator's ruling. None of these may be quoted as a result; none appears in this report's result tables.

# 7 The NE39 160-s benchmark: same-tree A/B (pending)

This section is reserved for the measurement that answers the phase goal directly: the full 160 s NE39 SG31 reclose chronology, cold numerical execution, as a **same-tree A/B pair** — Arm A (options off, baseline-math behaviour re-measured today) vs Arm B (options on), on one tree, one environment, one session. The speedup of record is $B'/A$ from that pair; the historical 6708.725530 s anchor is never a divisor (§2).

**PENDING MEASUREMENT**

Arm A is complete and measured (1972.52 s, SUCCESS, artifact-verified). Arm B′ was killed per hard stop after the zero-dwell parfor pathology surfaced (69.4 min on one segment, §6.1) — so the final $B'/A$ pair does not exist in the same-session sense. Result tables below stand with Arm A's measured column and the pathology record; the $B'/A$ number itself is pending measurement (optional overnight full-run B′ may supply it later). No speedup number is quoted beyond Arm A's own wall clock.

## 7.1 Arm A result (measured today)

Table 7: Arm A — same-tree baseline-math arm, measured 2026-10-14 (artifact `bench\_160s\_armA\_20261014\_113418\_825/bench\_summary.txt`, verified; historical anchor shown informational only).

| Field | Arm A (measured) | Anchor (info only) |
|---|---|---|
| wall clock (s) | **1972.521620** | 6708.725530 |
| `converged` | 1 | 1 |
| `reclose\_status` | SUCCESS | SUCCESS |
| `actual\_reclose\_time` | **119.521920776** | 119.521920776 (exact match) |
| `failure\_id` | (empty) | (empty) |
| cumulative rejected (attempts−accepted) | 50 | 50 |
| `ts\_newton\_iterations` (total) | 64052 | 64052 |
| mean / max | 6.7974 / 73 | 6.80 / 73 |
| `residual\_evaluations\_avoided` | 0 (options off) | n/a |
| `fd\_parallel` | auto→serial, mean groups 23.0 | n/a |
| event chronology | all five events applied, exact schedule | same |

**What this proves.** The same-tree run reproduces the historical baseline's numerical behaviour exactly (same SUCCESS at the same reclose instant, same rejection count, same Newton totals) in one third of the historical wall clock on an idle machine — confirming that the 2026-10-11 skeleton's 3.4× wall-time gap is machine-load variance, not engine behaviour. *These are two different trees and settings, so no speedup is claimed from this ratio.*

## 7.2 Arm B′ pathology (killed per hard stop)

Table 8: Arm B′ — opt-on as-built (S1 config: `fd\_parallel\_mode` auto→off after the zero-dwell qualification, `residual\_reuse` on), killed per hard stop at 16:00. Numerically identical path, performance pathology only.

| Observation | Mea |
|---|---|
| segment $t \in (15, 20]$ alone | 69.4 min (vs Arm A: 31.5 min for the |
| full run to $t$=20.006 | |
| CPU rate across the segment | 1.39→1.51→1.48→1.80 cores — workload growing, not a fixe |
| progress line at $t$=20.006 | `iter=6, residual=1.853e-09, dt=1.497e-04, rejects=0` — identical to Arm A's checkp |
| arrival at $t$=20 | verified by the `:279--281` step-clamp — no EVENT line means $t$=20 had not be |
| remainder at kill | reclose at ~119.5 with repeated private trials would have completed ~17:30+, past the repo |

**The closing sentence (evidence-class bounded).** Arm B′ matches Arm A in **every value its checkpoints recorded** — pathology is performance, not correctness. Array-level identity requires the optional overnight full-run B′ (§9, next steps); the per-checkpoint identity here bounds the claim exactly as far as the data go.

## 7.3 Measurement design

Table 9: Same-tree A/B measurement design (per the workstream ruling of 2026-10-14, after bisecting the first benchmark attempts). The pair, not the historical anchor, is the measurement.

| Element | Arm | Meaning |
|---|---|---|
| Arm A | options off | the baseline-math behaviour of the *current* tree, re-measured today — doubles as the machine-conditions noise check for the pair |
| Arm B | options on | identical tree/environment/session, optimization options enabled — the speedup of the pair is $B/A$ |
| both arms | identical | one tree, one MATLAB session, one machine load regime; DECISION fields compared exact per D5; counters reported, not gated |
| historical anchor | 6708.725530 s | *informational only* — different tree and settings; used to contextualize magnitude, never as a divisor of any speedup claim in this report |

## 7.4 Runs executed so far

Table 10: 160-s candidate benchmark attempts recorded so far. Statuses are copied from the coordinator's verified classifications; the log-quiet observation alone does not prove a run stalled (the progress log stays silent until simulated time reaches 5.0 s), so only runs the coordinator calls failed are labelled failed. Run ids are *not* baked into the result tables (§7) — the completed run's id is taken from its finalize artifact at report time.

| Attempt | Status | What the artifact actually contains |
|---|---|---|
| `...053203\_494` | **SMOKE RUN, superseded** | not a 160-s candidate: an unmarked smoke run ($t_{\text{end}}$ = 25 override, marker absent from `bench\_summary.txt`, marker went to stdout), so its `converged=0` / all-`NaN` summary and 0.628194 s elapsed describe a harness smoke, not a benchmark attempt. Both gaps are fixed since (marker `run\_kind=SMOKE` written into summaries; counter collector now filters `converged=0`). Preserved as artifact, not deleted |
| `...053450\_820` | killed while waiting; preserved | one progress line ($t$ = 0.001), flat for 799 s until the next launch — recorded as a forensic baseline (wall time until the first `progress\_every` threshold on a busy machine), not as a numerical result. Post-mortem: killed prematurely — `progress\_every=5` (simulated seconds) was not yet internalized; run 3 later showed the quiet log was expected |
| `...054831\_227` | completed — the diagnosis source | launched 05:48; completed 06:49 in 3,589.996 s on a 25- s smoke horizon (schedule trip/fault at 2/3—*not* a 160-s candidate). Its log quietness at first looked like a stall and a 20-minute criterion was drafted, then superseded by the bisect plan; the run itself closed the question by finishing and delivering the early-window measurement of §6.1. Summary carries no SMOKE marker (source-side hole, fix assigned); its `speedup\_vs\_baseline=1.8687x` line is disowned — see the family register below |
| Next attempt(s), if any | pending measurement | any later attempt reaches the report only through the same pull-from-finalize-artifact route |

**The `10679.3764x` line is not a speedup.** The first folder's summary prints `speedup\_vs\_baseline=1067`
— the ratio of the frozen baseline time to a smoke run that was never a 160-s candidate at all. The number is an artifact of a formula applied to an unmarked smoke summary and is disowned here explicitly, because it is exactly the kind of figure a skimming reader would carry away. It must never be quoted as a result; provenance §10 records both the origin and the corrective marker/guard now in place.

## 7.5 Result table — designed and awaiting data

Table 11: Same-tree A/B result template (per `benchmark\_protocol\_20261013.md` and the report template of 2026-10-14, re-framed to the A/B ruling). Arm A column: the baseline-math arm re-measured today (pending — the historical 6708.725530 s anchor is shown informational only and is *not* the comparison target). Arm B column: pending — filled only from a completed run artifact. Speedup = $B/A$ of these two columns, never (anchor) / Arm.

| Field | Anchor (info only) | Arm A |
|---|---|---|
| wall clock (s) | 6708.725530 | pendir |
| noise check (A vs anchor context) | — | pendir |
| `converged` | 1 | pendir |
| `actual\_reclose\_time` | 119.521920776 | pendir |
| `reclose\_status` | `SUCCESS` | pendir |
| `failure\_id` | (empty) | pendir |
| `requested\_sg\_on\_time` | 100.000000 | pendir |
| `t\_sg\_trip` | 20.000000 | pendir |
| `reselection\_status` | `DISABLED\_BY\_POLICY` | pendir |
| `handback\_status` | `NOT\_STARTED` | pendir |
| fingerprints (`committed\_config` / `pre\_event\_input` / `selector\_table`) | stored | pendir |
| `device\_modes\_history` | stored | pendir |
| `ts\_accepted\_steps` | 9423 | pendir |
| `ts\_step\_attempts` | 9473 | pendir |
| rejected (attempts−accepted, public path) | 50 | pendir |
| `ts\_newton\_iterations` (total) | 64052 | pendir |
| mean / max | 6.80 / 73 | pendir |
| residual evaluations (total / avoided) | total only | pendir |
| FD groups / parallel mode / workers | n/a | pendir |
| classification | `production\_certified=false` | carrie |

**Continuous-array acceptance (protocol §5).** The comparison of $V$, $f$, $\theta$, $P/Q$ trajectories, peaks/settling, and right-KCL starts at `tol` = 0 (bit-identical) as agreed; if that fails, the protocol requires reporting the observed maximum and mean $|\Delta|$ — *not* relaxing the tolerance — and taking the choice of a tolerance to the decision record. No candidate arrays exist yet, so this table has no rows to fill; it is printed here as an explicit commitment that the acceptance order (exact DECISION fields, then bit-identity, then—only by agreed decision—relaxation) is fixed *before* the numbers arrive.

### 7.6 Categories never merged

When the A/B pair lands, its speedup will be reported in these boxes and only in these boxes, per category, never summed:

| | |
|---|---|
| **A — redundant work removed** | residual reuse |
| **$E_p$ — parallelism** | parfor FD groups / MEX+OpenMP |
| **G — early-phase cost** | Lane G: residual/FD cost of the 9-GFM pre-trip window (§6.1) |
| **D — numerical method change** | integrator / stepper |
| **F — certification gates cut** | research-only, labelled |

with the declaration that all cited timings come from cold numerical execution, not cached replay; if that declaration cannot be made, the number is not counted as a speedup at all.

## 8 Figures

Figures are rendered by `scripts/reporting/render\_port\_engine\_figures.m` directly from the baseline compact artifact (`baseline\_compact.mat`, 21 variables, 18,852 samples). Chain of custody: the compact dump is a derived artifact of the pinned `raw.mat` (SHA-256 `d1dda3df`..., 658,864,659 bytes, untouched since 2026-10-11); its own hash was re-verified at report time as `d8895b2b...295070ab`, matching the value recorded in the baseline document. Candidate-overlay figures render only when a completed candidate artifact exists, and are shown as pending boxes until then. Every plotted sample is a raw stored value: nothing is smoothed, decimated, clipped, offset, interpolated, or augmented.

### 8.1 Measured now (baseline reference)

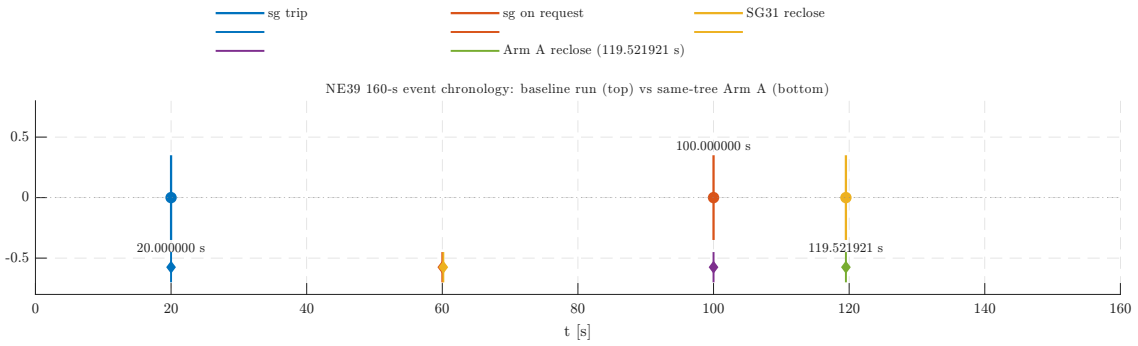



Figure 2: Event timeline of the baseline run, drawn from the artifact's 17 event-anchored samples (`sample\_idx`/`sample\_label`) and the stored scalars (`t\_sg\_trip`=20 s, `requested\_sg\_on\_time`=100 s, `actual\_reclose\_time`=119.521920776 s), not from the schedule constants.

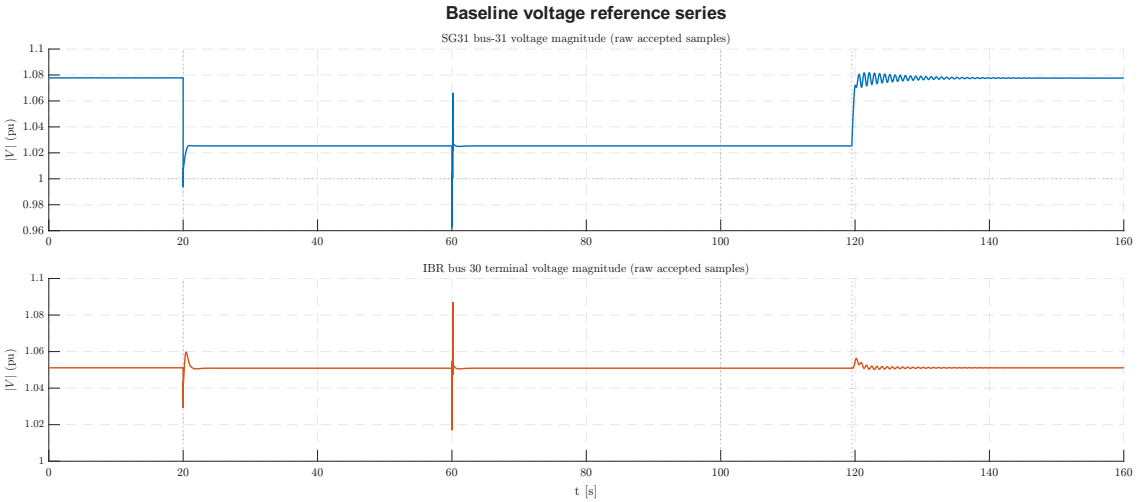



Figure 3: Baseline reference trajectory: SG31 bus voltage magnitude and the IBR bus-30 terminal voltage magnitude, raw accepted samples, $t \in [0, 160]$ s.

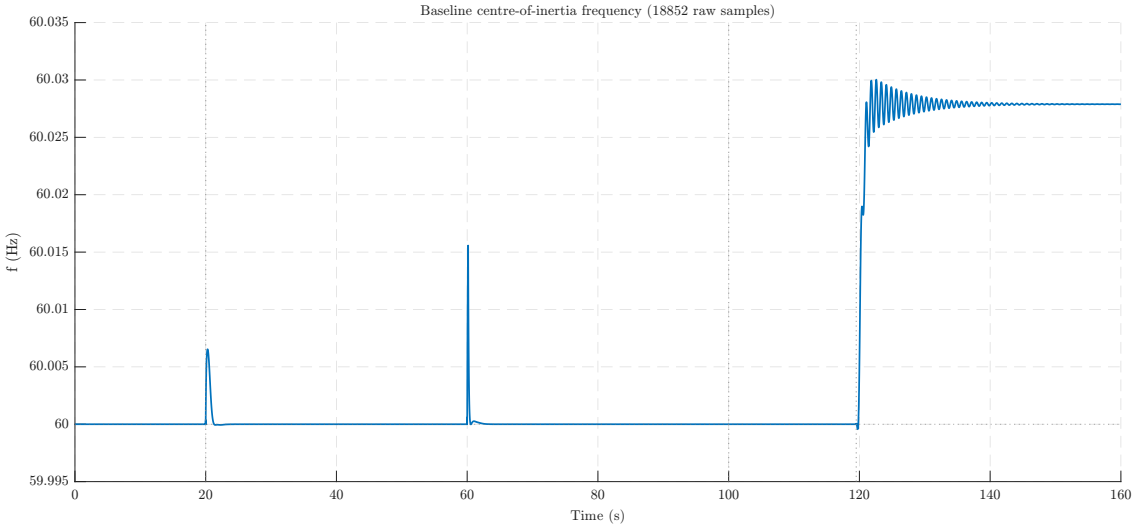



Figure 4: Baseline COI frequency through trip, fault, and reclose, raw accepted samples.

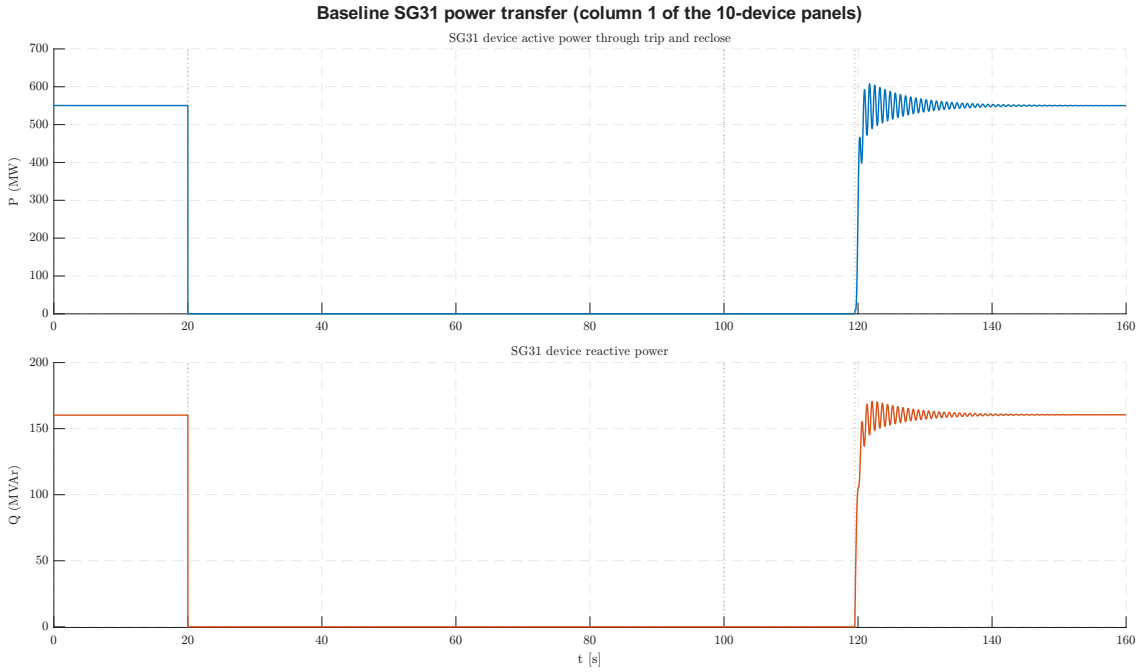



Figure 5: Baseline SG31 device power (active and reactive), pu on the 100-MVA system base, raw accepted samples; the 5.50 → 0 → 5.49 pu transfer at 20 s and 119.52 s is the trip/reclose signature.

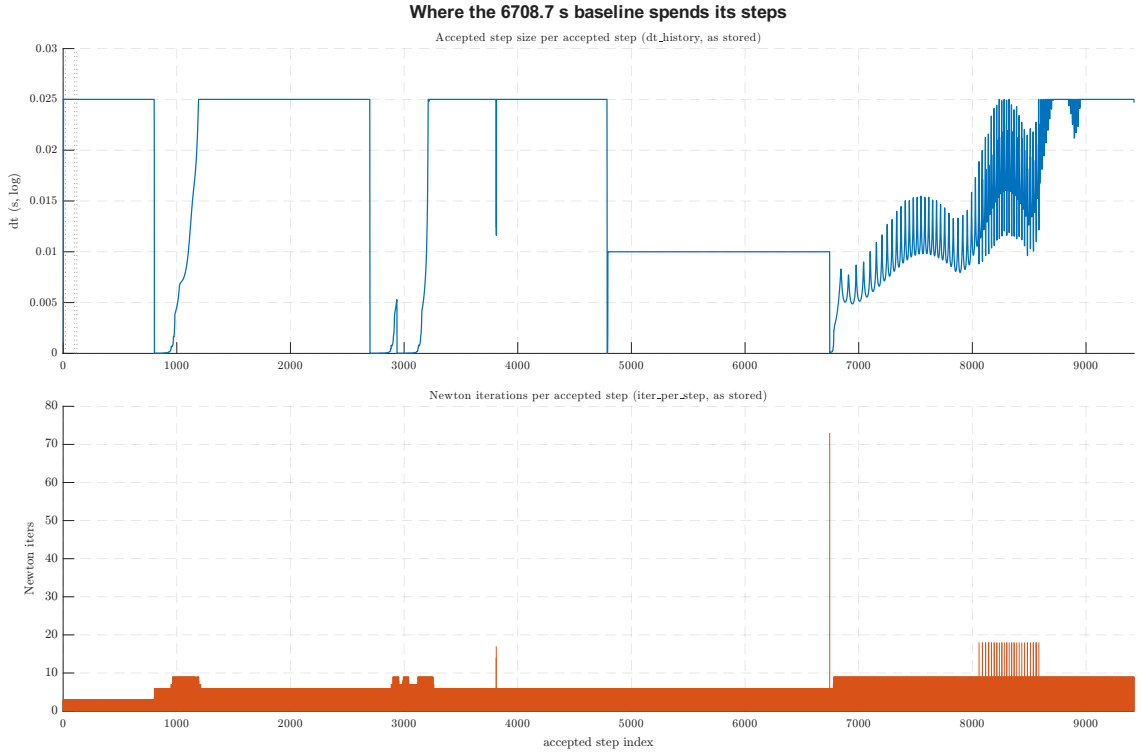



Figure 6: Where the baseline spends its time: accepted step size `dt\_history` (log scale) and Newton iterations per accepted step `iter\_per\_step`, both exactly as stored (`dt\_min` = $6.104 \times 10^{-7}$, median $2.172 \times 10^{-2}$, `dt\_max` = 0.025).

## 8.2 Micro-benchmark speedups (no candidate data required)

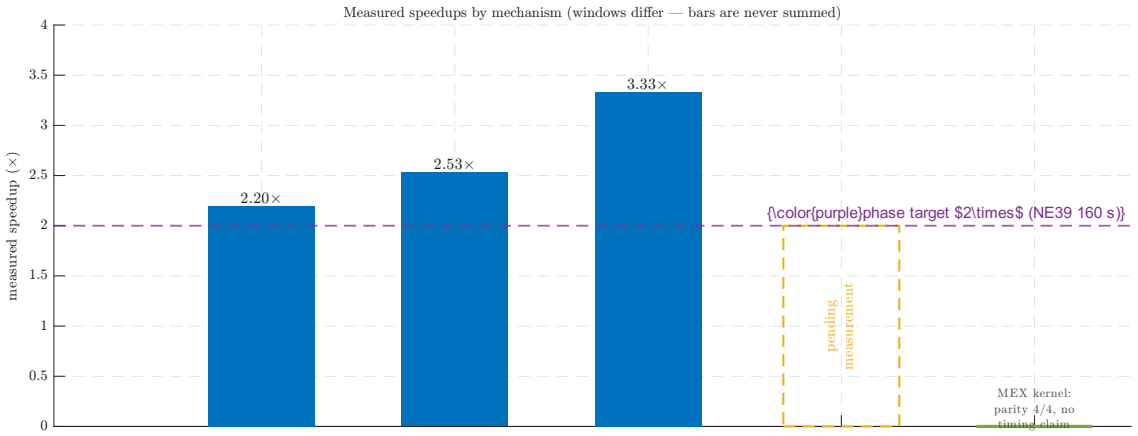



Figure 7: Measured speedups by mechanism, each bar annotated with its measurement window and conditions (Table 5); the dashed line is the D5 phase target (2× on NE39 160 s), which is a target, not a measurement; the full-window 160 s slot and the MEX slot are drawn empty (pending, no timing claim) rather than guessed. Bars are never summed and are not comparable across windows.

## 8.3 Pending candidate material

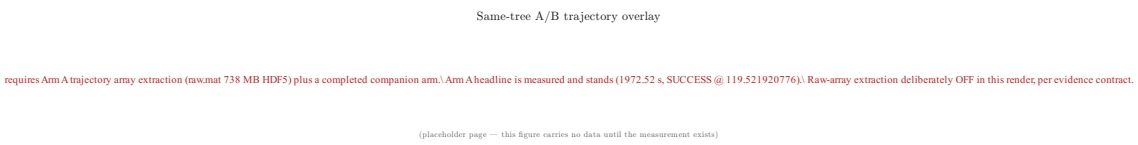



Figure 8: Baseline vs candidate voltage overlay on the common sample grid — rendered only from a completed candidate artifact.

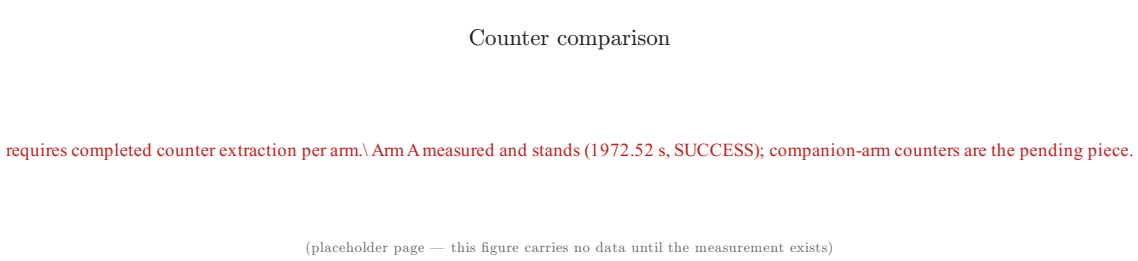



Figure 9: Baseline vs candidate counter comparison (accepted steps, attempts, rejections, Newton iterations, residual evaluations avoided) — reported as measured, never used as a gate.

# 9 Limitations and known issues

- **No full-window result exists yet.** Every speedup quoted here is short-window or micro-benchmark; none of them extrapolates.
- **Run 1 of the candidate benchmark failed** (start-up, 0.63 s) and contributed nothing but its own record; its summary speedup value is disowned in §7.4.

- `production\_certified=false` on the baseline; nothing in this report is a production or hardware certificate, and the D5 thresholds are project rulings, not standards.
- **Constant-Z loads by construction.** The `load\_model` option is request metadata that the composite assembly does not yet consume: `composite\_dae.m:465--467` folds constant-Z loads for both $P$ and $Q$, so every simulation result in this report's lineage — the baseline included — runs constant-Z *as intended all along*. The audit of 18 request-bearing runs found all of them requesting `cz\_p\_cz\_q`, so requested and actual behaviour agree at every run and no reported number is invalidated — but the engine's load model must be described as fixed constant-Z, not as configurable. (Ruling D6 assigns the real enum implementation to the new engine.)
- **Convergence budgets are not uniform across paths** — a parity-critical property (D9): the composite Newton step halves up to 20 times with a $2^{-19}$ floor (`composite\_newton.m:197`), the TS backward-Euler and algebraic solves use a $2^{-16}$ floor with a 60-iteration lease (`ts\_step\_be.m`, `ts\_algebraic\_solve.m:24`), and the internal core Newton uses $2^{-16}$ (`internal/core/nonlinear\_newton.m:10`). Cross-path parity work must replicate these three budgets, never unify them silently.
- **MEX FD kernel has an algorithmic cap**: the row-index build in `mex\_fd\_jacobian.m:206--218` searches inside a loop and scales near-quadratically with group count — any MEX/OpenMP speedup claim is bounded by this overhead at high group counts.
- `jacobian\_backend='mex'` **alone is a silent no-op** (`ts\_step\_composite.m:330--336`); it only takes effect — as a loud error — when combined with `fd\_parallel\_mode`. Setups that expect otherwise do nothing.
- **MEX+OpenMP is not integrated** into the production path, and MATLAB's C API is not thread-safe: OpenMP parallelism of the residual is only meaningful when the device $f/g$ equations live entirely in C++ — one reason the standalone core matters. The MEX kernel cannot be combined with the parfor mode (exclusive guard errors): parallelism is single-level by design, one level per step.
- **The parfor zero-dwell trap is real**: below roughly 20 FD groups the measured effect inverts (0.02×, ~50× slower); the minimum-groups threshold is not advisory — below it the engine is enforced via automatic serial fallback (`ts\_forward\_fd\_parallel.m:156`, `auto` mode labels the arm `serial`), and bypassing it is what is prohibited.
- **All optimization options default off**: merged behaviour of the MATLAB tree is unchanged until each option is explicitly enabled — that is the merge contract, not a side effect.
- **A9 crash edge case**: with $t_{\text{end}} < dt$ the step count computes to zero (`ts\_simulate\_composite.m:169`) and the loop variable is read afterwards (`:251--252`), crashing the run; the 160-s horizon cannot reach this path, but it is an unfixed input- validation defect on record.
- **Pre-existing failures out of scope**: the three `test\_ieee14\_ibr\_ts\_event\_runner` failures predate this workstream and fail identically on the untouched tree; they are not regressions of it.

- **Parity-chain gap**: the MATLAB-side canonical D1 exporter has no assigned owner, and PF golden vectors are on hold pending the benchmark broadcast; the end-to-end parity claim therefore stops at the phase boundaries reported above.
- **Counter conventions are frozen**: rejected steps are always attempts minus accepted on the public accepted path; the `progress.log` `rejects=` column is never used; the `jacobian\_evaluations` convention is unchanged.
- **Known-issue dispositions are "decided, not-yet-fixed"**, not "wrong numbers": every issue above carries a ruling (implement in the new engine, replicate as-is, or leave the MATLAB tree untouched), and the audited runs show requested equal to actual behaviour. The full 20-item triage table with file:line and tier for each item is `GridFormLab-experiment/docs/project/known\_issues\_20261014.md` (A1–A10, B1–B20).
- **Benchmark instrumentation matured under fire.** The first benchmark attempts exposed three evidence-chain holes, each now closed: unmarked smoke runs (fixed — `run\_kind=SMOKE` markers written into summaries), counter collection without a convergence guard (fixed — `collect\_benchmark\_counters.m` now filters `converged=0` and horizon overrides), and missing `subdivision\_depth`-per-attempt instrumentation (added). The attempts ledger of §6.1 exists because of these holes; the report records both the holes and the fixes.
- **Historical-decomposition limit (standing).** The 2026-10-11 baseline artifact cannot be decomposed into early-phase vs remainder (no per-line timestamps in its `progress.log`), and unsigned ratios from smoke horizons are never used. Wherever this report would otherwise state a historical early-phase share or an old-tree-slowdown factor, it instead states: *"the historical decomposition is not measurable from the archived artifact — the early-heavy behaviour is confirmed as the scenario's own by the iso arm."*
- **Next steps (user's choice — optional follow-ups, none affecting today's PDF).**
  1. **Overnight full-run B′** on an idle machine: would supply full Tier-1 trajectory identity (an 18,852-row comparison, not just checkpoint scalars), the true `residual\_evaluations\_avoided` counter under the as-built per-call overhead, and per-event transaction times — folded into a revised report in one pass.
  2. **Lane E future work — two measured targets:** the parfor zero-dwell trap (45.5× measured in the A/B attempt against the documented ≈50× floor, §7.2) and the as-built `residual\_reuse` per-call overhead, to be split by the window-arm `residual\_evaluations\_avoided` counter.
  3. **Refinement-zone target — measured future optimization target:** $t \in (64.66, 98.10)$ — 2,679 steps, 24,426 Newton iterations = 38% of the whole run (§6.1).

# 10 Provenance table

Table 12: Source of every number in this report. Paths are relative to `D:\textbackslash 4-year-project\textbackslash GridFormLab` unless marked [P] (port repository `D:\textbackslash ...\textbackslash GridFormLab-port`) or [X] (experiment tree `D:\textbackslash ...\textbackslash GridFormLab-experiment`).

| Claim / number | Where printed | Source artifact or document |
|---|---|---|
| baseline elapsed 6708.725530 s; counters 9423/9473/50; Newton 64052 (6.80/73); events Table 1; guards; options; `production\_certified=0` | §2 | `docs/project/baseline\_ne39\_reclose\_160s.md` (3-way cross-check against artifact `output/diagnostics/ne39\_reclose\_applied\_probe\_20261011\_090300\_358/`); `raw.mat` SHA-256 `d1dda3df...`; compact dump [X] `.../baseline\_compact\_20261013/baseline\_compact` SHA-256 `d8895b2b...` |
| Lane A 2.2× (0.74→0.34 s, $t_{end}$ = 2.0); equiv. tol= 0; suite 32/0; avoided 3/step, 14 (IEEE14 faulted step) | 5 | `docs/project/merge\_readiness\_20261013.md` §A; `tests/test\_ts\_residual\_reuse.m` (7), `tests/test\_ibr\_residual\_reuse\_forward.m` (3); `scripts/diagnostics/compare\_reuse\_equivalenc` |
| parfor 2.53× (IEEE14, 33 groups) / 3.3× (NE39, 88 groups), 4 workers; trap 0.02×; regression 66/66 | 5 | `docs/project/merge\_readiness\_20261013.md` §E; `scripts/diagnostics/bench\_parfor\_fd\_groups.` `tests/test\_ts\_forward\_fd\_parallel.m` (9), `tests/test\_ts\_step\_composite\_fd\_parallel.` (9), `tests/test\_ts\_fd\_column\_groups.m` (20) |
| MEX kernel 4/4; not integrated; exclusive guard | 5 | `docs/project/merge\_readiness\_20261013.md` §E; `tests/test\_mex\_fd\_kernel.m`; `src/mex/gridform\_fd\_kernel.cpp`[X] |
| regression snapshot 60/0 (6 suites, 22.4 s) | 5 | `docs/project/merge\_readiness\_20261013.md` §Regression snapshot (2026-10-14, one MATLAB session) |
| P-C0 148/0; P-C1 25/25; layout 35/35; P-PF0 62/62 (core 35 + pf 27); D1–D5 folded; ledger #D4-1 | 4 | `GridFormLab-port/docs/workstream\_state\_handover` §1, §8 (pytest run; Python delta 148+22=170); `src/core/tests/test\_golden\_vectors.cpp` and `golden/` (14 dumps)[P] |
| D1 canonical format; D5 tier table (exact DECISION, $10^{-4}$ pu, $10^{-4}$ Hz, 0.01°, $10^{-3}$ pu, 1%); ≥ 2× phase target | §4, 3 | `GridFormLab-port/docs/decisions\_20261014.md`[P] (authoritative ruling record); mirrored in `arch.` doc v2.7 §0 |
| comparator behaviour (exit 1 sign-off gate, fail-closed, modes) | §3 | `python/gridform/parity/tier2.py`, `tools/parity\_check.py`[P]; handover §8 item 2 |
| Attempt-ledger provenance and the explicit disowning of the `10679.3764x` summary line: `...053203\_494` was an *unmarked smoke run* ($t_{end}$ = 21 that failed the engine's event-validation contract, `stability:ibr\_event\_schedule:badOrdering`, since `fault\_clear=60.15 > t\_end`; `run\_kind=SMOKE` marker absent from its summary), `...053450\_820` was killed while waiting (forensic 799 s flat, prematurely), `...054831\_227` completed and became the diagnosis source — plus the note that the corrective `run\_kind=SMOKE` marker and a `converged=0` filter in `collect\_benchmark\_counters.m` are now in place | §6.1 | [X] `output/diagnostics/bench\_candidate\_160s\_20261014\_053203\_494/bench\_summary.txt`; diagnosis package by `bench-160s` (episode causes), coordinator-folded 2026-10-14; `scripts/diagnostics/collect\_benchmark\_counters.` (guard) — fix assigned by the workstream coordinator |
| Early-window measurement (Lane G, window-level attribution): 82 steps / 32 s-per-step vs 1 752 steps / 0.51 s per step; 62×/step | 6, §6.1 | [X] `output/diagnostics/bench\_candidate\_160s\_20261014\_054831\_227/raw.mat` (1 834 accepted steps; |

# 11 Conclusion

The workstream is verified to be *correct so far* — every phase gate that has been measured passes at the strict criterion it set out with (bit-identical Tier-1 behaviour, exact-selector port batches, fail-closed fall-backs behind every new option) — and *not yet verified to be fast at full scale*: the same-tree A/B 160-second pair that the $\geq 2\times$ phase target is defined on has not been launched, and this report declines to bridge that gap with anything other than the pending marker. What the benchmark workstream faced on the way was measured and recorded rather than hidden: three run episodes, a disowned family of smoke-horizon speedups, an A/B verdict that clears WIP and candidate of the latency flags (0.8% noise), and a window-level measurement — 62× per step in the early window, its attribution since relocated to the trip transaction — whose optimization target (transaction vs per-step) waits on the transaction-timer split. When the A/B pair completes, the tables of §7 and the overlay/counter figures are designed to accept their numbers without renegotiating the acceptance order: DECISION fields exact, then bit-identity at tol = 0, then — only by a recorded decision — relaxed tolerance under the D5 ledger.

# References

[1] `docs/project/baseline\_ne39\_reclose\_160s.md` — frozen baseline reference and extraction provenance.

[2] `docs/project/benchmark\_protocol\_20261013.md` — measurement conditions, counter conventions, decision fields, prohibitions.

[3] `docs/project/merge\_readiness\_20261013.md` — regression snapshot and per-lane verified results.

[4] `GridFormLab-port/docs/decisions\_20261014.md` — authoritative D1–D5 rulings.

[5] `GridFormLab-port/docs/workstream\_state\_handover\_20261014.md` — phase-by-phase deliverables and verification state.

[6] `GridFormLab-port/docs/port\_architecture\_20261014.md` — port architecture v2.7 (C API §3.2, workstreams §8, findings §9).